

# Vasilios Nicholas

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Available: January 2027 – December 2027 | GitHub: [github.com/vasiliosnicholas](https://github.com/vasiliosnicholas) | Personal Site: [vasilios-nicholas.com](https://vasilios-nicholas.com)

## EDUCATION

**Northeastern University** – Boston, MA Aug 2025 – Present  
*MS in Computer Science Candidate, GPA: 4.0 / 4.0* *Expected Graduation: December 2027*  
Relevant Coursework: Algorithms; Object-Oriented Design; Data Structures, Algorithms and their Applications within Computer Systems;  
Web Development; Master's Research; Computer Systems (in progress);

**Johns Hopkins University** – Baltimore, MD Aug 2020 – Aug 2021  
*MS in Healthcare Management* Aug 2016 – May 2020  
*BS in Neuroscience, Degree Conferred with General Honors*

## TECHNICAL KNOWLEDGE

**Languages:** Python, Java, C, C++, JavaScript, TypeScript, HTML5, CSS3, Bash, LaTeX, Markdown

**Tools, Frameworks:** Git, GDB, NumPy, Pandas, SciPy, Matplotlib, JUnit, React.js, Svelte, SvelteKit, Node.js, Express.js, MongoDB, Docker, Podman

**Other Systems:** Unix/Linux, Windows 11, eClinicalWorks, Athenahealth, ImageTrend, Microsoft Office, Claude Code, GitHub

## PROJECTS

**Pupillometry Filtering Package** | Python, TypeScript, Svelte | *Northeastern SATH Lab, Massachusetts General Hospital* Jan 2026 – Present  
• Developing an open-source pupillometry noise filtering and analysis package as a Khoury Master's Research Apprentice and [Research Assistant](#).  
• [Presented](#) the blink detection and filtering algorithm at the Khoury Master's Research Symposium on April 14<sup>th</sup>, 2026  
• Currently developing a web app for researchers to clean their pupillometry data using the package.

**Agility — Agile Project Management Platform** | TypeScript, React, Node.js, Express, MongoDB | *Northeastern University* July – Aug 2026  
• Built the authentication and team-management stack of an Agile project management application, deployed on Render with MongoDB Atlas.  
• Implemented session-based authentication with Passport Local and bcrypt, enforcing role-based route guards across the API and SPA.  
• Developed the team management, ticket assignment, and profile CRUD flows with React Hook Form and Yup; added arrow-key navigation.

**NavHealth — Healthcare Facility Search Engine** | JavaScript (ES6), Node.js, Express, MongoDB | *Northeastern University* June – July 2026  
• Built the reviews subsystem of a search application that surfaces healthcare facilities by service, insurance accepted, and distance.  
• Implemented a **REST API** with full CRUD over a MongoDB reviews collection, sorted by recency, rating, or popularity.  
• Rendered review interfaces client-side with vanilla JavaScript fetch calls, over a database seeded from Massachusetts facility profile datasets.

**CS 5004 Object-Oriented Design — Adventure Game** | Java | *Northeastern University* Nov – Dec 2025  
• Designed and implemented a text- and GUI-based (**Swing**) adventure game using **Model View Controller** architecture and **SOLID** principles.  
• Wrote **JUnit** tests covering the model and controller modules to verify each component's functionality.  
• Modeled the class hierarchy in **Unified Modeling Language (UML)** before implementation, designing for readability and extensibility, including using a single controller to handle both the text- and GUI-based view modules.

## WORK EXPERIENCE

**Research Assistant Intern** | Massachusetts General Hospital – Boston, MA May 2026 – Present  
• Developing an **open-source Python** package that detects and filters blink artifacts from raw pupillometry recordings, enabling collaborating clinical researchers to evaluate the kinematics of pupil diameter as a biomarker of circadian system health.  
• Implemented a **blink-filtering algorithm** and validated output against manually labeled data to confirm physiological signal is preserved.  
• **Scaled** the algorithm to clean **840+** study signals reducing total subsequent manual cleaning time **to under 2 hours**.  
• **Evaluated linear and non-linear regression models** using goodness-of-fit metrics to identify the optimal model for quantifying pupil kinematics.  
• Designed the **API and documentation** for clinical researchers, eliminating a critical, time-intensive data-cleaning bottleneck.  
• Developing a **front-end app** for researchers to clean and view analysis on their pupillometry data using the package in **Svelte and TypeScript**.

**Graduate Teaching Assistant** | Northeastern University, Khoury College – Boston, MA Jan 2026 – Apr 2026  
• Led a weekly recitation and biweekly office hours for a **DSA** course for graduate students entering CS from non-technical backgrounds.  
• Used recurring errors identified during grading to restructure recitation content, targeting the conceptual gaps the class had rather than those assumed by the syllabus.  
• Developed comprehensive **C test suites** for each assignment to expose edge cases in students' **DSA implementations** and provide actionable feedback.  
• Graded assignments on correctness, algorithmic complexity, and code readability, giving written feedback to **20 students weekly**.

**Clinical Research Coordinator** | The Institute for Advanced Reconstruction – Tinton Falls, NJ Jun 2022 – Aug 2023  
• Analyzed clinical datasets for prospective and retrospective studies, computing descriptive and inferential statistics that supported 3 manuscripts and conference abstracts.  
• Maintained **IRB compliance** across 2 active clinical studies, managing protocol documentation, consent records, and data integrity; trained an incoming coordinator on the same workflows.  
• **Authored** literature reviews and evidence-based insurance appeals that reversed coverage denials for reconstructive surgery patients.

**Data Analyst** | Seaview Orthopaedics and Medical Associates – Ocean Township, NJ Mar 2021 – Oct 2021  
• Collected and analyzed patient engagement for the practice to identify bottlenecks in scheduling and follow-up workflows.  
• Partnered with administration and IT to evaluate, select, and deploy new patient engagement resulting in increased patient portal adoption.

## LEADERSHIP AND VOLUNTEERING

**Holmdel First Aid Squad** – Holmdel, NJ | Volunteer EMT-B and Crew Chief Jan 2019 – Aug 2023  
• **Led** crews of 2-4 across approximately **400 annual 911 responses**, assessing, treating, and transporting patients, escalating to advanced life support when required, assisting with motor vehicle extrication, and assigning crew roles and assessing crew performance under time pressure.

## HONORS, AWARDS, AND CERTIFICATIONS

**Honors:** Spring 2026 Khoury Master's Research Apprenticeship (Northeastern University), Dean's List (Johns Hopkins University, Fall 2016, Spring 2017, Fall 2019); Nu Rho Psi National Neuroscience Honor Society (2019); Eagle Scout (2016)

**Certifications:** NJ Emergency Medical Technician (EMS ID 638304, expired); AHA Basic Life Support Provider; PEARS Provider

**Languages:** English (fluent); Greek (intermediate), German (intermediate)